

Student ID:40735842

11. Find the order in the database which has the wrong delivery charge applied

SELECT

o.id AS order\_id,

ROUND(SUM(fi.price \* oi.quantity), 2) AS items\_total,

CASE

WHEN SUM(fi.price \* oi.quantity) < 10 THEN 3

WHEN SUM(fi.price \* oi.quantity) >= 10 AND SUM(fi.price \* oi.quantity) < 20 THEN 2

WHEN SUM(fi.price \* oi.quantity) >= 20 AND SUM(fi.price \* oi.quantity) < 30 THEN 1

WHEN SUM(fi.price \* oi.quantity) >= 30 THEN 0

END AS delivery,

ROUND(SUM(fi.price \* oi.quantity), 2) +

CASE

WHEN SUM(fi.price \* oi.quantity) < 10 THEN 3

WHEN SUM(fi.price \* oi.quantity) >= 10 AND SUM(fi.price \* oi.quantity) < 20 THEN 2

WHEN SUM(fi.price \* oi.quantity) >= 20 AND SUM(fi.price \* oi.quantity) < 30 THEN 1

WHEN SUM(fi.price \* oi.quantity) >= 30 THEN 0

END AS total,

o.delivery\_charge AS stored\_delivery\_charge,

o.total\_price AS stored\_total

FROM orders o

JOIN order\_item oi ON o.id = oi.order\_id

JOIN food\_item fi ON oi.food\_item\_id = fi.id

GROUP BY o.id, o.delivery\_charge, o.total\_price

HAVING

(items\_total < 10 AND o.delivery\_charge != 3)

OR (items\_total >= 10 AND items\_total < 20 AND o.delivery\_charge != 2)

OR (items\_total >= 20 AND items\_total < 30 AND o.delivery\_charge != 1)

OR (items\_total >= 30 AND o.delivery\_charge != 0);

```
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->      WHEN SUM(fi.price * oi.quantity) >= 20 AND SUM(fi.price * oi.quantity) < 30 THEN 1  
->      WHEN SUM(fi.price * oi.quantity) >= 30 THEN 0  
->      END AS total,  
->      o.delivery_charge AS stored_delivery_charge,  
->      o.total_price AS stored_total  
-> FROM orders o  
-> JOIN order_item oi ON o.id = oi.order_id  
-> JOIN food_item fi ON oi.food_item_id = fi.id  
-> GROUP BY o.id, o.delivery_charge, o.total_price  
-> HAVING  
->      (items_total < 10 AND o.delivery_charge != 3)  
->      OR (items_total >= 10 AND items_total < 20 AND o.delivery_charge != 2)  
->      OR (items_total >= 20 AND items_total < 30 AND o.delivery_charge != 1)  
->      OR (items_total >= 30 AND o.delivery_charge != 0);  
+-----+-----+-----+-----+-----+-----+  
| order_id | items_total | delivery | total | stored_delivery_charge | stored_total |  
+-----+-----+-----+-----+-----+-----+  
|      89625 |          30.00 |          0 | 30.00 |              3.00 |          33.00 |  
+-----+-----+-----+-----+-----+-----+  
1 row in set (1.762 sec)
```

12. How are orders distributed over the day and over the week? Count all orders starting on the hour and before the start of the next hour?

SELECT \* FROM (

SELECT

DAYNAME(o.order\_date) AS day,

SUM(CASE WHEN HOUR(o.order\_date) = 11 THEN 1 ELSE 0 END) AS `1100`,

SUM(CASE WHEN HOUR(o.order\_date) = 12 THEN 1 ELSE 0 END) AS `1200`,

SUM(CASE WHEN HOUR(o.order\_date) = 13 THEN 1 ELSE 0 END) AS `1300`,

SUM(CASE WHEN HOUR(o.order\_date) = 14 THEN 1 ELSE 0 END) AS `1400`,

SUM(CASE WHEN HOUR(o.order\_date) = 15 THEN 1 ELSE 0 END) AS `1500`,

SUM(CASE WHEN HOUR(o.order\_date) = 16 THEN 1 ELSE 0 END) AS `1600`,

SUM(CASE WHEN HOUR(o.order\_date) = 17 THEN 1 ELSE 0 END) AS `1700`,

SUM(CASE WHEN HOUR(o.order\_date) = 18 THEN 1 ELSE 0 END) AS `1800`,

SUM(CASE WHEN HOUR(o.order\_date) = 19 THEN 1 ELSE 0 END) AS `1900`,

SUM(CASE WHEN HOUR(o.order\_date) = 20 THEN 1 ELSE 0 END) AS `2000`,

SUM(CASE WHEN HOUR(o.order\_date) = 21 THEN 1 ELSE 0 END) AS `2100`

FROM orders o

GROUP BY DAYNAME(o.order\_date)

UNION ALL

SELECT

'All days' AS day,

SUM(CASE WHEN HOUR(o.order\_date) = 11 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 12 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 13 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 14 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 15 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 16 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 17 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 18 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 19 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 20 THEN 1 ELSE 0 END),

SUM(CASE WHEN HOUR(o.order\_date) = 21 THEN 1 ELSE 0 END)

FROM orders o

) AS summary

ORDER BY

FIELD(day, 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday',  
'All days');

```

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-> SUM(CASE WHEN HOUR(o.order_date) = 18 THEN 1 ELSE 0 END),
-> SUM(CASE WHEN HOUR(o.order_date) = 19 THEN 1 ELSE 0 END),
-> SUM(CASE WHEN HOUR(o.order_date) = 20 THEN 1 ELSE 0 END),
-> SUM(CASE WHEN HOUR(o.order_date) = 21 THEN 1 ELSE 0 END)
-> FROM orders o
-> ) AS summary
-> ORDER BY
-> FIELD(day, 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday', 'All days');
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| day   | 1100 | 1200 | 1300 | 1400 | 1500 | 1600 | 1700 | 1800 | 1900 | 2000 | 2100 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Monday | 177  | 1455 | 1381 | 431  | 110  | 72   | 298  | 967  | 2291 | 1619 | 844  |
| Tuesday | 220  | 1779 | 1683 | 533  | 135  | 91   | 354  | 1142 | 2722 | 1946 | 1011 |
| Wednesday | 224  | 1798 | 1714 | 542  | 133  | 88   | 303  | 986  | 2356 | 1667 | 864  |
| Thursday | 163  | 1329 | 1268 | 401  | 102  | 67   | 305  | 999  | 2379 | 1700 | 877  |
| Friday  | 176  | 1433 | 1348 | 426  | 108  | 71   | 387  | 1257 | 3002 | 2121 | 1099 |
| Saturday | 47   | 353  | 341  | 104  | 25   | 18   | 357  | 1179 | 2777 | 1981 | 1019 |
| Sunday  | 24   | 186  | 174  | 61   | 15   | 9    | 328  | 1081 | 2568 | 1825 | 949  |
| All days | 1031 | 8333 | 7909 | 2498 | 628  | 416  | 2332 | 7611 | 18095 | 12859 | 6663 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
8 rows in set (0.170 sec)

```

13. One post code is responsible for more orders than any other. Show the customers from that post code, the number of orders they have each made and their average daily spend to the nearest penny?

```

SELECT
  a.post_code,
  u.first_name,
  u.last_name,
  COUNT(o.id) AS orders,
  ROUND(SUM(o.total_price) / DATEDIFF(MAX(o.order_date), MIN(o.order_date)), 2) AS
  average_daily_spend
FROM orders o
JOIN address a ON a.id = o.delivery_address_id
JOIN users u ON u.id = o.customer_id
WHERE a.post_code = (
  SELECT post_code
  FROM address
  GROUP BY post_code
  ORDER BY COUNT(*) DESC
  LIMIT 1

```

)

GROUP BY a.post\_code, u.first\_name, u.last\_name

ORDER BY orders DESC;

```
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-> JOIN users u ON u.id = o.customer_id  
-> WHERE a.post_code = (  
->   SELECT post_code  
->   FROM address  
->   GROUP BY post_code  
->   ORDER BY COUNT(*) DESC  
->   LIMIT 1  
-> )  
-> GROUP BY a.post_code, u.first_name, u.last_name  
-> ORDER BY orders DESC;  
  
+-----+-----+-----+-----+-----+  
| post_code | first_name | last_name | orders | average_daily_spend |  
+-----+-----+-----+-----+-----+  
| EH3 9LT | Bailey | Davidson | 45 | 55.97 |  
| EH3 9LT | Maleah | Chase | 43 | 55.61 |  
| EH3 9LT | Sylvia | Moon | 41 | 52.24 |  
| EH3 9LT | Logan | Randall | 40 | 63.14 |  
| EH3 9LT | Jabari | Barnes | 38 | 47.36 |  
| EH3 9LT | Marcus | Hopkins | 37 | 51.25 |  
| EH3 9LT | Cali | Richard | 36 | 58.22 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.193 sec)  
  
MariaDB [40735842]>
```

14. Who are the people who have placed entirely vegetarian orders and who share the same first and last name?

WITH vegetarian\_orders AS (

SELECT DISTINCT o.customer\_id

FROM orders o

JOIN order\_item oi ON o.id = oi.order\_id

JOIN food\_item fi ON oi.food\_item\_id = fi.id

WHERE fi.vegetarian = 1

AND NOT EXISTS (

SELECT 1

FROM order\_item oi2

JOIN food\_item fi2 ON oi2.food\_item\_id = fi2.id

WHERE oi2.order\_id = o.id AND fi2.vegetarian = 0

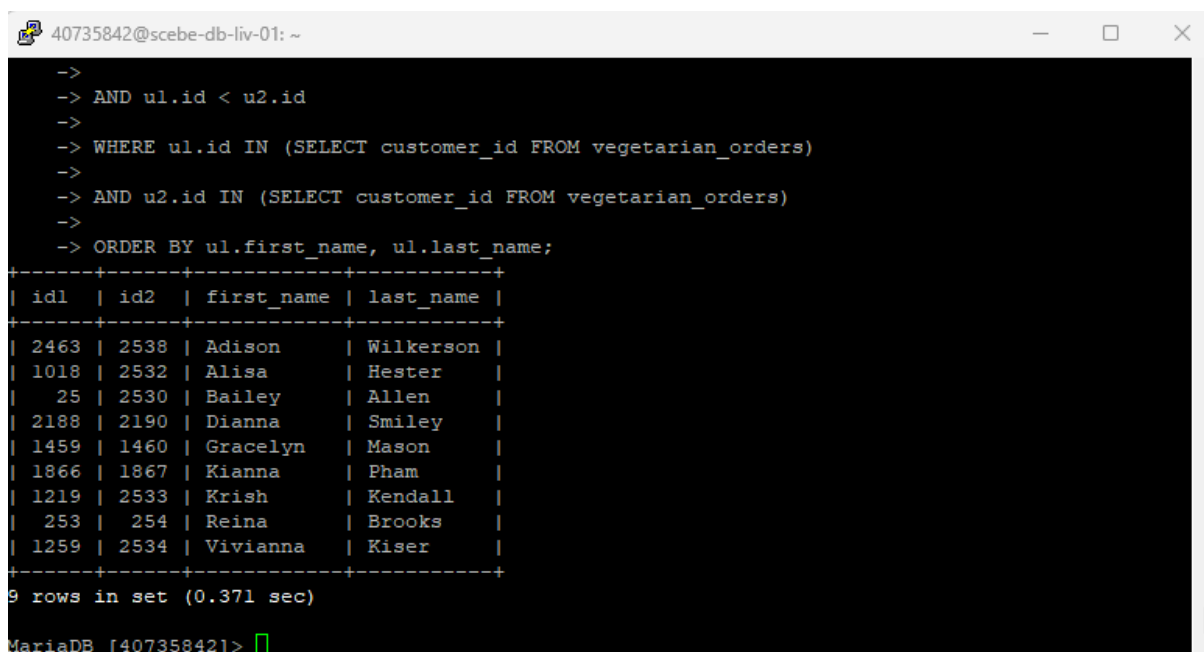
)

```

)

SELECT
u1.id AS id1,
u2.id AS id2,
u1.first_name,
u1.last_name
FROM users u1
JOIN users u2 ON u1.first_name = u2.first_name
AND u1.last_name = u2.last_name
AND u1.id < u2.id
WHERE u1.id IN (SELECT customer_id FROM vegetarian_orders)
AND u2.id IN (SELECT customer_id FROM vegetarian_orders)
ORDER BY u1.first_name, u1.last_name;

```



```

40735842@scebe-db-liv-01: ~
->
-> AND u1.id < u2.id
->
-> WHERE u1.id IN (SELECT customer_id FROM vegetarian_orders)
->
-> AND u2.id IN (SELECT customer_id FROM vegetarian_orders)
->
-> ORDER BY u1.first_name, u1.last_name;
+-----+
| id1 | id2 | first_name | last_name |
+-----+
| 2463 | 2538 | Adison     | Wilkerson |
| 1018 | 2532 | Alisa      | Hester    |
| 25   | 2530 | Bailey     | Allen     |
| 2188 | 2190 | Dianna     | Smiley    |
| 1459 | 1460 | Gracelyn   | Mason     |
| 1866 | 1867 | Kianna     | Pham      |
| 1219 | 2533 | Krish      | Kendall   |
| 253   | 254   | Reina      | Brooks    |
| 1259 | 2534 | Vivianna  | Kiser     |
+-----+
9 rows in set (0.371 sec)

MariaDB [40735842]>

```

15. If I order a delivery between 1830 and 1930 on a Friday, what is the probability it will arrive within 40 minutes (exclusive, <40)?

```

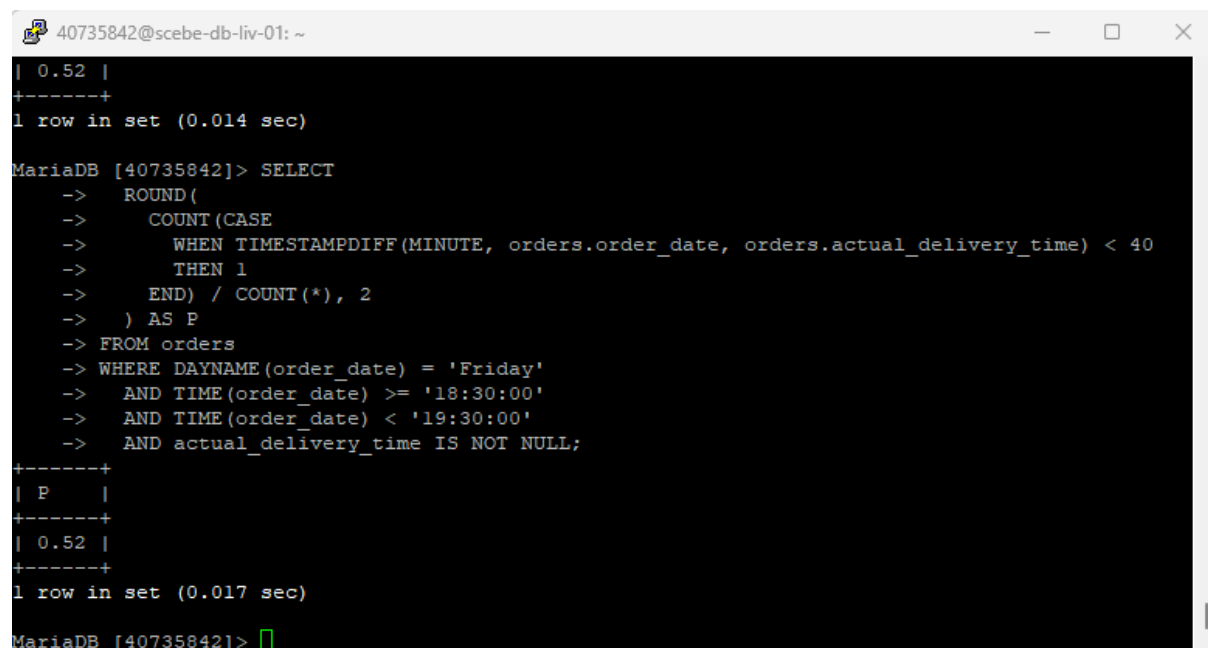
SELECT

```

```

ROUND(
COUNT(CASE
    WHEN TIMESTAMPDIFF(MINUTE, orders.order_date, orders.actual_delivery_time) <
40
    THEN 1
END) / COUNT(*), 2
) AS P
FROM orders
WHERE DAYNAME(order_date) = 'Friday'
AND TIME(order_date) >= '18:30:00'
AND TIME(order_date) < '19:30:00'
AND actual_delivery_time IS NOT NULL;

```



```

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| 0.52 |
+-----+
1 row in set (0.014 sec)

MariaDB [40735842]> SELECT
-> ROUND(
->   COUNT(CASE
->     WHEN TIMESTAMPDIFF(MINUTE, orders.order_date, orders.actual_delivery_time) < 40
->     THEN 1
->   END) / COUNT(*), 2
-> ) AS P
-> FROM orders
-> WHERE DAYNAME(order_date) = 'Friday'
->   AND TIME(order_date) >= '18:30:00'
->   AND TIME(order_date) < '19:30:00'
->   AND actual_delivery_time IS NOT NULL;
+-----+
| P      |
+-----+
| 0.52   |
+-----+
1 row in set (0.017 sec)

MariaDB [40735842]> █

```